

Emdadul Haque Apu *Software Engineer*

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Proficient in programming, database systems, and technical problem-solving through professional software engineering experience. Eager to contribute to research projects involving data collection, literature review, experimentation, and scientific reporting.

PROFESSIONAL EXPERIENCE

Full Stack Developer L2, *Manush AI Platform LTD*

Aug 2025 – Present

- Designed, developed, and delivered 300+ REST APIs and 50+ backend modules supporting large-scale enterprise workflows
- Optimized PostgreSQL performance through indexing, query tuning, and materialized views, reducing report generation times by up to 10x for datasets containing tens of millions of records
- Designed relational database schemas and data models using Prisma and PostgreSQL, ensuring data integrity and maintainable application architecture
- Optimized complex SQL queries and implemented Redis-based locking mechanisms to prevent race conditions and handle concurrency edge cases.
- Designed and implemented automated background processes, scheduled tasks, and data synchronization workflows using 20+ cron jobs
- Implemented JWT, Redis, and cookie-based authentication and authorization systems, improving application security and session management
- Built and maintained CI/CD pipelines to automate testing, deployment, and release workflows, reducing manual deployment effort and improving delivery reliability.
- Built responsive frontend applications using React, TypeScript, Next.js, Mantine UI, Redux Toolkit, and Tanstack Query, improving usability and development efficiency
- Led feature delivery, code reviews, technical estimation, and cross-team collaboration while mentoring developers in Agile environments

Jr. Software Engineer, *TechHub, Save the Children International*

Jan 2023 – Aug 2025

- Built a React-NestJS based SaaS CMS (*SCI React Admin*) and successfully delivered 3+ client projects using the platform.
- Developed and maintained RESTful APIs for 10+ applications using Node.js, improving backend performance by 30%.
- Built Node.js (Express) middlewares to optimize MongoDB queries, reducing response time by 40%
- Improved application performance by 40–80% by minimizing data fetching and implementing efficient state management
- Developed interactive and reusable UI components with React, Vue, and Next in 7+ applications, improving development efficiency and maintaining UI consistency
- Collaborated with the Data Analytics team to deliver 2+ PowerApps solutions that streamlined internal processes and improved operational efficiency.

EDUCATION

BSc in Electronics and Communication Engineering (ECE), *National University*

Academic and Co-curricular Activities

- Actively participated in inter-dept programming contests
- Successfully demonstrated and defended the final year project on Low-Cost Spin Coater
- Contributed to multiple outsourced web projects contractually

SKILLS

Language — JavaScript, TypeScript, SQL, C++

Backend — Node.js, NestJS, Express.js, REST APIs

Databases/ORMs — PostgreSQL, MongoDB, Redis, Prisma, TypeORM

Frontend — React, Next.js, Redux Toolkit, React Hook Form, Formik

Styling/UI — Tailwind CSS, Material UI, Mantine UI, Shadcn UI, Bootstrap

DevOps/Cloud — Docker, Git, CI/CD, AWS S3

Concepts — Microservices, System Design, OOP

PROJECTS

Prism, *Field Force Automation Platform (FMCG, Bangladesh)*

• *NestJS* • *React* • *PostgreSQL* • *Prisma* • *Mantine UI* • *Redux Toolkit* • *Tanstack Query* • *ExcelJS*

- Delivered 10+ full-stack features and mobile APIs enabling real-time operations with cross-functional teams
- Developed and optimized backend services integrating NestJS, PostgreSQL and Redis for high-performance data management.
- Generated materialized views and applied necessary schema tuning for analytics over billions of records
- Built cron-based scheduling systems for automated reporting, data sync, and backend workflows

SCI-React-Admin, *Custom CMS Platform (3+ projects delivered)*

• *NestJS* • *React* • *PostgreSQL* • *TypeORM* • *Material UI* • *React Hook Form* • *Redux Toolkit* • *RTK Query* • *Axios*

- Developed a full-stack CMS using React, NestJS, and PostgreSQL with modular architecture
- Designed and optimized database schema and queries using TypeORM for scalable data access
- Implemented role-based access control (RBAC) for secure content and user management
- Reduced redundant API calls and improved performance by up to 80% through caching and optimization

COURSES

Introduction to DevOps — IBM, Coursera

Fundamentals of CI/CD, IaC, automated deployments, microservices, and DevOps culture.

Introduction to Containers w/ Docker, Kubernetes & OpenShift — IBM, Coursera

Fundamentals of containerization, Kubernetes architecture, and YAML-based deployment of cloud-native applications.

LANGUAGES

English — Professional proficiency

Bangla — Native Proficiency

CERTIFICATIONS

Duolingo English Test (Score — 125, CEFR Level — B2–C1)